# CAPSTONE PROJECT

# Customer Churn Prediction System Using Machine Learning: An End-to-End Data Science Capstone Project

## 1. Project Overview and Objectives

### Project Overview

This project predicts customer churn using features such as tenure, monthly charges, contract type, and payment method. The workflow includes data preprocessing, customer segmentation, and machine learning modeling. A Random Forest classifier is used to identify churn patterns and predict high-risk customers. The system helps businesses understand churn drivers and implement targeted retention strategies.

### Objectives

- To analyze customer data to identify patterns and key factors influencing churn behavior.
- To preprocess and prepare the dataset for effective machine learning modeling.
- To develop and evaluate a machine learning model for predicting customer churn.
- To build a local application that allows users to run churn predictions using the trained model.
- To generate actionable insights that help businesses improve customer retention strategies.

## 2. Setup and Installation Instructions

### Prerequisites

- Python 3.10+
- Jupyter Notebook

### Installation Steps

1. Download or clone the project repository from GitHub.

2. Install required libraries:

   pip install -r requirements.txt

3. Place dataset inside /data/raw folder.

4. Launch Jupyter Notebook and open the notebook:

   capstone_project.ipynb

## 3. Code Structure Explanation

```
comprehensive_ds_project_week12/
│
├── capstone_project.ipynb
│
├── data/
│   ├── raw/
│   │   └── customer_churn.csv
│   └── processed/
│       └── cleaned_data.csv
│
├── src/
│   ├── data_loader.py
│   ├── preprocessing.py
│   ├── feature_engineering.py
│   ├── model_training.py
│   ├── evaluation.py
│
├── deployment/
│   ├── app.py
│   └── model.pkl
│
├── reports/
│   ├── model_report.md
```

```
│       └── business_report.md
├── presentation/
│       └── capstone_presentation.pptx
├── requirements.txt
└── README.md
```

**File Descriptions**

- data/raw/customer_churn.csv

  Original dataset containing customer attributes such as tenure, monthly charges, contract type, payment method, and churn labels (0 = retained, 1 = churned).

- data/processed/cleaned_data.csv

  Preprocessed dataset after data cleaning, encoding categorical variables, and preparing features for machine learning models.

- capstone_project.ipynb

  Jupyter notebook containing the complete data science workflow including data exploration, preprocessing, feature engineering, model training, evaluation, and churn risk analysis.

- src/data_loader.py

  Utility functions for loading raw datasets and saving processed datasets used throughout the project pipeline.

- src/preprocessing.py

  Implements data preprocessing steps including handling missing values, encoding categorical variables, and preparing the dataset for modeling.

- src/feature_engineering.py

  Functions for creating the feature matrix and target variable and preparing model-ready inputs.

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- src/model_training.py
  Contains functions for training machine learning models, performing hyperparameter tuning, and saving the trained model.

- src/evaluation.py
  Provides functions for evaluating model performance using metrics such as accuracy, ROC-AUC, confusion matrix, and classification report.

- deployment/app.py
  Streamlit application that allows users to input customer data and run churn predictions through a web interface on localhost.

- deployment/model.pkl
  Serialized Random Forest machine learning model used by the application to generate churn predictions.

- reports/model_report.md
  Technical report summarizing the model development process, training methods, evaluation metrics, and feature importance results.

- reports/business_report.md
  Business-focused report explaining churn drivers, insights from data analysis, and recommendations for reducing customer churn.

- presentation/capstone_presentation.pptx
  Project presentation summarizing the business problem, methodology, model performance, and key insights.

- requirements.txt
  Specifies all Python dependencies needed to run the project.

- README.md
  Project documentation including overview, setup instructions, and usage guide.

## 4. Screenshots of Working Application

The included screenshots demonstrate:

- The Customer Churn Distribution plot indicates a class imbalance, with significantly more non-churn customers than churn customers.
- The Contract Type Analysis chart shows that customers with month-to-month contracts exhibit the highest churn rate, while long-term contracts are associated with stronger customer retention.
- The Monthly Charges vs Churn boxplot reveals that customers with higher monthly charges tend to have greater churn probability, indicating possible price sensitivity.
- The Tenure vs Churn analysis highlights that customers with shorter tenure are more likely to churn, whereas long-tenure customers demonstrate strong loyalty.
- The Feature Correlation heatmap identifies Tenure as the most negatively correlated feature with churn (-0.51), suggesting longer customer relationships significantly reduce churn risk.
- The Model evaluation results show strong predictive performance, with Random Forest achieving 96% accuracy and ROC-AUC of 0.99, outperforming Logistic Regression.
- The High-Risk Customer analysis successfully identifies customers with churn probabilities exceeding 80%, enabling proactive retention strategies.

```
INFO:src.evaluation:Model evaluation completed
Accuracy: 0.96
ROC AUC: 0.9877425944841676
              precision    recall  f1-score   support

           0       0.96      1.00      0.98        89
           1       1.00      0.64      0.78        11

    accuracy                           0.96       100
   macro avg       0.98      0.82      0.88       100
weighted avg       0.96      0.96      0.96       100
```

Fig. 4.1. Model Evaluation Metrics

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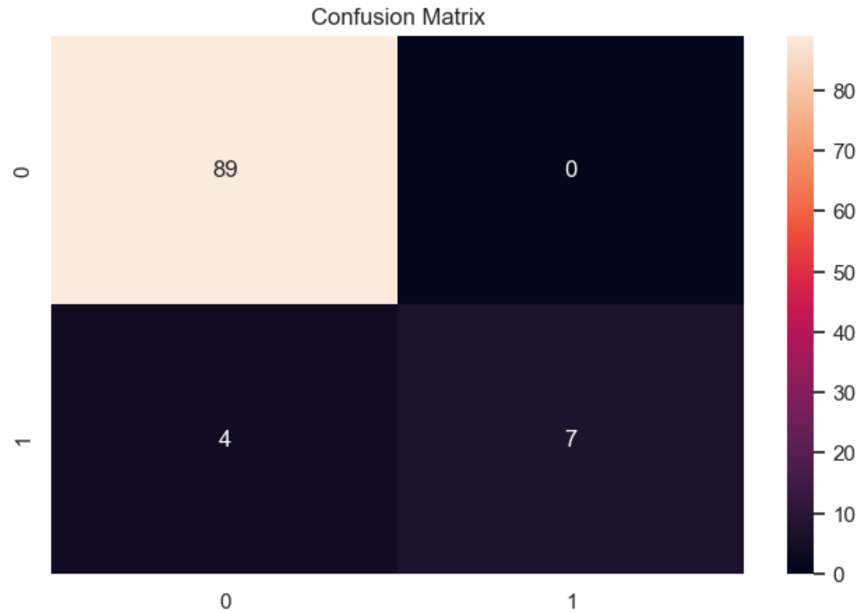


Fig. 4.2. Confusion Matrix

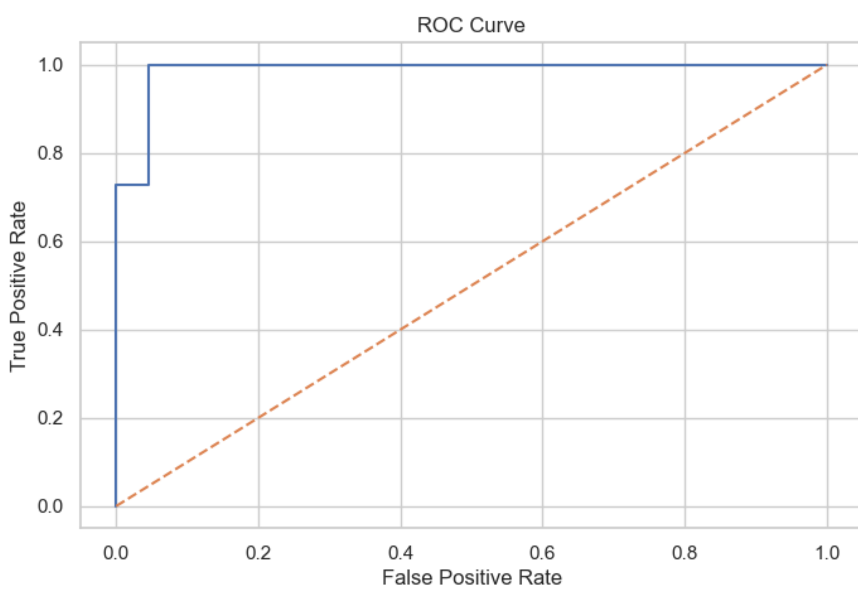


Fig. 4.3. ROC Curve

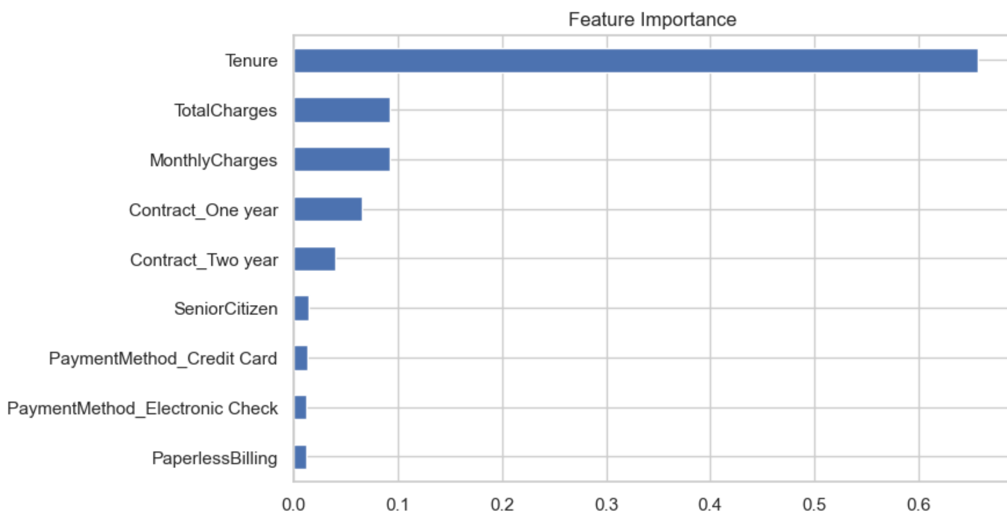


Fig. 4.4. Feature Importance

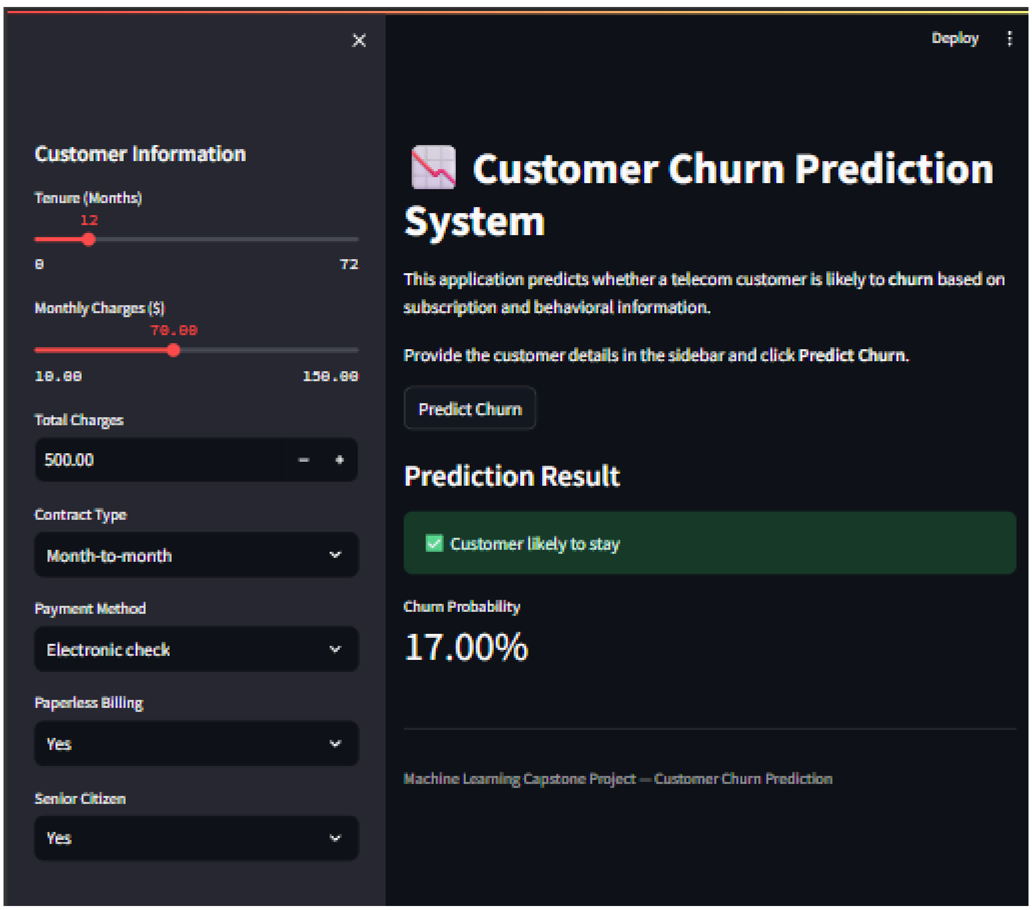


Fig. 4.5. Streamlit UI

The results show that the implemented ML pipeline effectively predicts churn and provides useful insights for telecom customer retention.

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## 5. Technical Requirements Fulfillment Explanation

| Category | Technical Requirement | How It Was Implemented |
|---|---|---|
| Algorithm | Data loading and preprocessing | Customer churn dataset loaded using Pandas; irrelevant identifier (CustomerID) removed; categorical variables (Contract, PaymentMethod, PaperlessBilling) encoded using one-hot encoding. |
| Algorithm | Data exploration | Dataset structure inspected using .head(), .info(), and .describe(); churn distribution and feature relationships visualized using Seaborn plots. |
| Algorithm | Feature preparation | Boolean features converted to numeric format and dataset validated to ensure all model inputs are numeric before training. |
| Algorithm | Model pipelines | Machine learning pipelines created using Scikit-learn to combine preprocessing and classification algorithms. |
| Algorithm | Baseline model comparison | Logistic Regression and Random Forest models trained and evaluated to compare baseline performance. |
| Algorithm | Hyperparameter tuning | Random Forest model optimized using GridSearchCV to tune parameters such as n_estimators and max_depth. |
| Algorithm | Model evaluation | Model performance evaluated using Accuracy, Precision, Recall, F1-Score, and ROC-AUC metrics. |

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| **Category** | **Technical Requirement** | **How It Was Implemented** |
|---|---|---|
| Algorithm | Feature importance analysis | Random Forest feature importance scores analyzed to identify the most influential factors affecting churn prediction. |
| Data Structure | Tabular data processing | Customer dataset stored and manipulated using Pandas DataFrames for preprocessing, feature engineering, and modeling. |
| Data Structure | Feature matrices | Input features extracted as feature matrix **X** and target variable **y** for supervised machine learning tasks. |
| Architecture | Modular ML workflow | Project implemented using modular Python scripts (data_loader, preprocessing, feature_engineering, model_training, evaluation). |
| Architecture | Reproducible data pipeline | Raw dataset stored in data/raw/; cleaned dataset saved to data/processed/ to maintain reproducibility across runs. |
| Architecture | Model training pipeline | End-to-end workflow includes preprocessing, feature engineering, model training, hyperparameter tuning, and evaluation. |
| Output | Visualization outputs | Generated visualizations including churn distribution charts, feature relationship plots, correlation heatmap, ROC curve, and feature importance plots. |

| Category | Technical Requirement | How It Was Implemented |
|---|---|---|
| Output | Model artifacts | Final trained Random Forest model saved as .pkl file using joblib for deployment and reuse. |
| Output | Prediction system | Model used to generate churn probability scores and identify high-risk customers for retention strategies. |
| Output | Business insights | Analysis revealed strong relationships between churn and contract type, tenure, and monthly charges, supporting targeted retention actions. |

## 6. Learning Outcomes

Through this project, the following learning outcomes were achieved:

- Gained hands-on experience in building an end-to-end machine learning pipeline, from data preprocessing to model evaluation.
- Performed exploratory data analysis and visualization to understand customer behavior and churn patterns.
- Developed and compared Logistic Regression and Random Forest models for churn prediction.
- Evaluated model performance using accuracy, precision, recall, F1-score, and ROC-AUC metrics.
- Identified key churn drivers using feature importance analysis to support business insights.
- Built a local web application to run churn predictions on localhost, demonstrating practical model deployment.

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## 7. Conclusion

This project developed a **Customer Churn Prediction System** using machine learning to identify customers likely to leave telecom services. Multiple models were evaluated, with **Random Forest achieving the best performance** in terms of accuracy and ROC-AUC. The system demonstrates an **end-to-end data science workflow** and helps businesses identify high-risk customers for targeted retention strategies.